CHAPTER

14

Nonlinearity and Mismatch

In Chapters 6 and 7, we dealt with two types of nonidealities, namely, frequency response and noise, that limit the performance of analog circuits. In this chapter, we study two other imperfections that prove critical in high-precision analog design and trade off with many other performance parameters. These effects are nonlinearity and mismatch.

We first define metrics for quantifying the effects of nonlinearity. Next, we study nonlinearity in differential circuits and feedback systems and examine several linearization techniques. We then deal with the problem of mismatch and dc offsets in differential circuits. Finally, we consider a number of offset cancellation methods and describe the effect of offset cancellation on random noise.

14.1 ■ Nonlinearity

14.1.1 General Considerations

As we have observed in the large-signal analysis of single-stage and differential amplifiers, circuits usually exhibit a nonlinear input/output characteristic. Depicted in Fig. 14.1, such a characteristic deviates from a straight line as the input swing increases. Two examples are shown in Fig. 14.2. In a common-source stage or a differential pair, the output variation becomes heavily nonlinear as the input level increases. In other words, for a small input swing, the output is a reasonable replica of the input, but for large swings the output exhibits "saturated" levels.

The nonlinear behavior of a circuit can also be viewed as *variation* of the slope, and hence the small-signal gain, with the input level. Illustrated in Fig. 14.3, this observation means that a given incremental change at the input results in different incremental changes at the output depending on the input dc level.

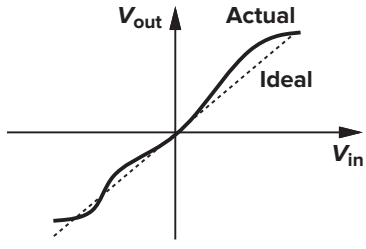


Figure 14.1 Input/output characteristic of a nonlinear system.

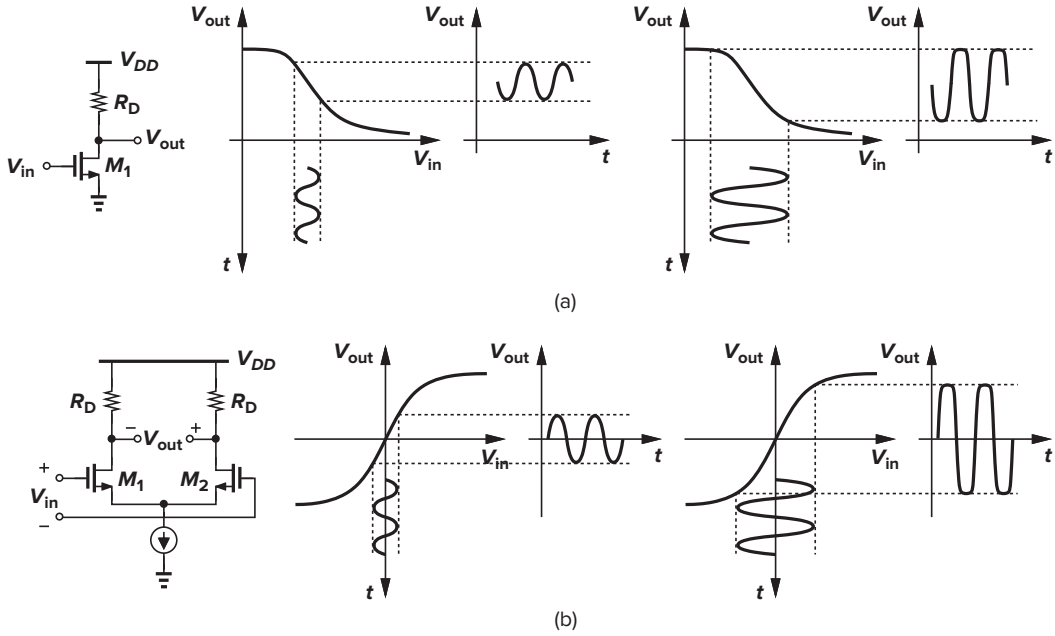


Figure 14.2 Distortion in (a) a common-source stage and (b) a differential pair.

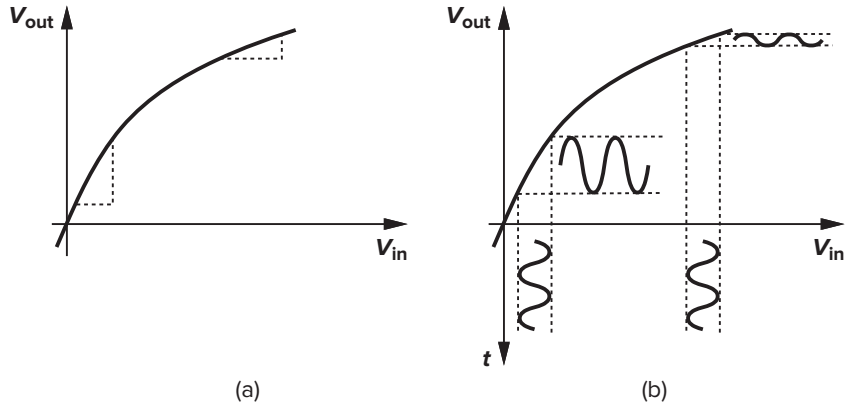


Figure 14.3 Variation of small-signal gain in a nonlinear amplifier.

In many analog circuits, precision requirements mandate relatively small nonlinearities, making it possible to approximate the input/output characteristic by a polynomial in the range of interest:

$$y(t) = \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t) + \cdots \tag{14.1}$$

For small x, $y(t) \approx \alpha_1 x$, indicating that α_1 is the small-signal gain in the vicinity of $x \approx 0$.

How is the nonlinearity quantified? A simple method is to identify α_1, α_2, etc., in (14.1). Another metric that proves useful in practice is to specify the maximum deviation of the characteristic from an ideal one (i.e., a straight line). As shown in Fig. 14.4, for the voltage range of interest, $[0 \;\; V_{in,max}]$, we pass a straight line through the end points of the actual characteristic, obtain the maximum deviation, ΔV, and normalize the result to the maximum output swing, $V_{out,max}$. For example, we say that an amplifier exhibits 1% nonlinearity ($\Delta V / V_{out,max} = 0.01$) for an input range of 1 V.

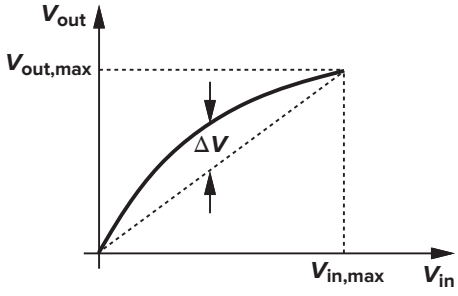


Figure 14.4 Definition of nonlinearity.

▶ Example 14.1

The input/output characteristic of a differential amplifier is approximated as $y(t) = \alpha_1 x(t) + \alpha_3 x^3(t)$. Calculate the maximum nonlinearity if the input range is from $x = -x_{max}$ to $x = +x_{max}$.

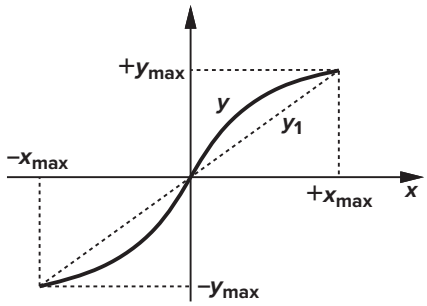


Figure 14.5

Solution

As depicted in Fig. 14.5, we can express the straight line passing through the end points as

$$y_1 = \frac{\alpha_1 x_{max} + \alpha_3 x_{max}^3}{x_{max}} x \tag{14.2}$$

$$= \left(\alpha_1 + \alpha_3 x_{max}^2\right) x \tag{14.3}$$

The difference between y and y_1 is therefore equal to

$$\Delta y = y - y_1 \tag{14.4}$$

$$= \alpha_1 x + \alpha_3 x^3 - \left(\alpha_1 + \alpha_3 x_{max}^2\right) x \tag{14.5}$$

Setting the derivative of Δy with respect to x to zero, we have $x = x_{max}/\sqrt{3}$, and the maximum deviation is equal to $2\alpha_3 x_{max}^3/(3\sqrt{3})$. Normalized to the maximum output, the nonlinearity is obtained as

$$\frac{\Delta y}{y_{max}} = \frac{2\alpha_3 x_{max}^3}{3\sqrt{3} \times 2\left(\alpha_1 x_{max} + \alpha_3 x_{max}^3\right)} \tag{14.6}$$

Note that the factor of 2 in the denominator is included because the maximum peak-to-peak output swing is equal to $2(\alpha_1 x_{max} + \alpha_3 x_{max}^3)$. For small nonlinearities, we can neglect $\alpha_3 x_{max}^3$ with respect to $\alpha_1 x_{max}$, arriving at

$$\frac{\Delta y}{y_{max}} \approx \frac{\alpha_3}{3\sqrt{3}\alpha_1} x_{max}^2 \tag{14.7}$$

Note that the relative nonlinearity is proportional to the square of the maximum input swing in this example. ◀

The nonlinearity of a circuit can also be characterized by applying a sinusoid at the input and measuring the harmonic content of the output. Specifically, if in (14.1), $x(t) = A\cos\omega t$, then

$$y(t) = \alpha_1 A\cos\omega t + \alpha_2 A^2\cos^2\omega t + \alpha_3\cos^3\omega t + \cdots \tag{14.8}$$

$$= \alpha_1 A\cos\omega t + \frac{\alpha_2 A^2}{2}[1+\cos(2\omega t)] + \frac{\alpha_3 A^3}{4}[3\cos\omega t + \cos(3\omega t)] + \cdots. \tag{14.9}$$

We observe that higher-order terms yield higher harmonics. In particular, even-order terms and odd-order terms result in even and odd harmonics, respectively. Note that the magnitude of the nth harmonic grows roughly in proportion to the nth power of the input amplitude. Called "harmonic distortion," this effect is usually quantified by summing the power of all of the harmonics (except that of the fundamental) and normalizing the result to the power of the fundamental. Such a metric is called the "total harmonic distortion" (THD). For a third-order nonlinearity,

$$\text{THD} = \frac{(\alpha_2 A^2/2)^2 + (\alpha_3 A^3/4)^2}{(\alpha_1 A + 3\alpha_3 A^3/4)^2} \tag{14.10}$$

Harmonic distortion is undesirable in most signal processing applications, including audio and video systems. High-quality audio products such as compact disc (CD) players require a THD of about 0.01% (−80 dB), and video products, about 0.1% (−60 dB).

14.1.2 Nonlinearity of Differential Circuits

Differential circuits exhibit an "odd-symmetric" input/output characteristic, i.e., $f(-x) = -f(x)$. For the polynomial of (14.1) to be an odd function, all of the even-order terms, α_{2j}, must be zero:

$$y(t) = \alpha_1 x(t) + \alpha_3 x^3(t) + \alpha_5 x^5(t) + \cdots \tag{14.11}$$

indicating that a differential circuit driven by a differential signal produces no even harmonics. This is another very important property of differential operation.

In order to appreciate the reduction of nonlinearity obtained by differential operation, let us consider the two amplifiers shown in Fig. 14.6, each of which is designed to provide a small-signal voltage gain of

$$|A_v| \approx g_m R_D \tag{14.12}$$

$$= \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})R_D \tag{14.13}$$

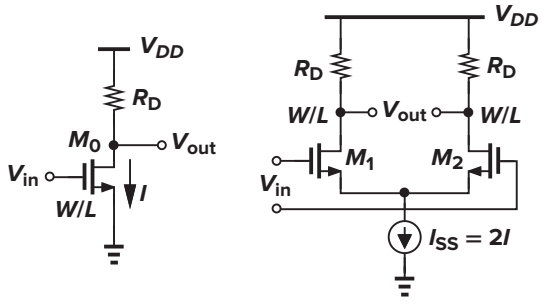


Figure 14.6 Single-ended and differential amplifiers providing the same voltage gain.

Suppose a signal $V_m \cos \omega t$ is applied to each circuit. Examining only the drain currents for simplicity, we can write for the common-source stage:

$$\begin{aligned} I_{D0} &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH} + V_m\, cos\omega t)^2 \\ &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2 + \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})V_m \cos\omega t \\ &\quad + \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_m^2 \cos^2\omega t \\ &= I + \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})V_m \cos\omega t + \frac{1}{4}\mu_n C_{ox}\frac{W}{L}V_m^2[1 + \cos(2\omega t)] \end{aligned} \tag{14.14}$$

Thus, the amplitude of the second harmonic, A_{HD2}, normalized to that of the fundamental, A_F, is

$$\frac{A_{HD2}}{A_F} = \frac{V_m}{4(V_{GS} - V_{TH})} \tag{14.15}$$

On the other hand, for M_1 and M_2 in Fig. 14.6, we have from Chapter 4

$$I_{D1} - I_{D2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{in}\sqrt{\frac{4I_{SS}}{\mu_n C_{ox}\dfrac{W}{L}} - V_{in}^2} \tag{14.16}$$

$$= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{in}\sqrt{4(V_{GS} - V_{TH})^2 - V_{in}^2} \tag{14.17}$$

If $|V_{in}| \ll V_{GS} - V_{TH}$, then

$$I_{D1} - I_{D2} = \mu_n C_{ox}\frac{W}{L}V_{in}(V_{GS} - V_{TH})\sqrt{1 - \frac{V_{in}^2}{4(V_{GS} - V_{TH})^2}} \tag{14.18}$$

$$\approx \mu_n C_{ox}\frac{W}{L}V_{in}(V_{GS} - V_{TH})\left[1 - \frac{V_{in}^2}{8(V_{GS} - V_{TH})^2}\right] \tag{14.19}$$

$$= \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})\left[V_m \cos\omega t - \frac{V_m^3 \cos^3\omega t}{8(V_{GS} - V_{TH})^2}\right] \tag{14.20}$$

Since $\cos^3 \omega t = [3\cos\omega t + \cos(3\omega t)]/4$, we obtain

$$I_{D1} - I_{D2} = g_m\left[V_m - \frac{3V_m^3}{32(V_{GS} - V_{TH})^2}\right]\cos\omega t - g_m\frac{V_m^3 \cos(3\omega t)}{32(V_{GS} - V_{TH})^2} \tag{14.21}$$

If $V_m \gg 3V_m^3/[8(V_{GS} - V_{TH})^2]$, then

$$\frac{A_{HD3}}{A_F} \approx \frac{V_m^2}{32(V_{GS} - V_{TH})^2} \tag{14.22}$$

Comparison of (14.15) and (14.22) indicates that the differential circuit exhibits much less distortion than its single-ended counterpart while providing the same voltage gain and output swing. For example, if $V_m = 0.2(V_{GS} - V_{TH})$, (14.15) and (14.22) yield a distortion of 5% and 0.125%, respectively.

While achieving a lower distortion, the differential pair consumes twice as much power as the CS stage because $I_{SS} = 2I$. The key point, however, is that even if the bias current of M_0 is raised to $2I$, (14.15) predicts that the distortion decreases by only a factor of $\sqrt{2}$ (with W/L maintained constant).

14.1.3 Effect of Negative Feedback on Nonlinearity

In Chapter 8, we observed that negative feedback makes the closed-loop gain relatively independent of the op amp's open-loop gain. Since nonlinearity can be viewed as variation of the small-signal gain with the input level, we expect that negative feedback suppresses this variation as well, yielding higher linearity for the closed-loop system.

Analysis of nonlinearity in a feedback system is quite complex. Here, we consider a simple, "mildly nonlinear" system to gain more insight. The reason is that, if properly designed, a feedback amplifier exhibits only small distortion components, lending itself to this type of analysis.

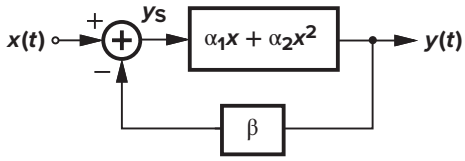


Figure 14.7 Feedback system incorporating a nonlinear feedforward amplifier.

Let us assume that the core amplifier in the system of Fig. 14.7 has an input-output characteristic $y \approx \alpha_1 x + \alpha_2 x^2$. We apply a sinusoidal input $x(t) = V_m \cos \omega t$, postulating that the output contains a fundamental component and a second harmonic and hence can be approximated as $y \approx a \cos \omega t + b \cos 2\omega t$.[1] Our objective is to determine a and b. The output of the subtractor can be written as

$$y_S = x(t) - \beta y(t) \tag{14.23}$$

$$= V_m \cos \omega t - \beta(a \cos \omega t + b \cos 2\omega t) \tag{14.24}$$

$$= (V_m - \beta a) \cos \omega t - \beta b \cos 2\omega t \tag{14.25}$$

This signal experiences the nonlinearity of the feedforward amplifier, thereby producing an output given by

$$\begin{aligned} y(t) &= \alpha_1[(V_m - \beta) \cos \omega t - \beta b \cos 2\omega t] \\ &\quad + \alpha_2[(V_m - \beta a) \cos \omega t - \beta b \cos 2\omega t]^2 \end{aligned} \tag{14.26}$$

$$\begin{aligned} &= [\alpha_1(V_m - \beta a) - \alpha_2(V_m - \beta a)\beta b] \cos \omega t \\ &\quad + \left[-\alpha_1 \beta b + \frac{\alpha_2(V_m - \beta a)^2}{2}\right] \cos 2\omega t + \cdots \end{aligned} \tag{14.27}$$

The coefficients of $\cos \omega t$ and $\cos 2\omega t$ in (14.27) must be equal to a and b, respectively:

$$a = (\alpha_1 - \alpha_2 \beta b)(V_m - \beta a) \tag{14.28}$$

$$b = -\alpha_1 \beta b + \frac{\alpha_2(V_m - \beta a)^2}{2} \tag{14.29}$$

[1] Note that higher harmonics and phase shifts through the system are neglected.

The assumption of small nonlinearity implies that both α_2 and b are small quantities, yielding $a \approx \alpha_1(V_m - \beta a)$ and hence

$$a = \frac{\alpha_1}{1 + \beta\alpha_1} V_m \tag{14.30}$$

which is to be expected because $\beta\alpha_1$ is the loop gain. To calculate b, we write

$$V_m - \beta a \approx \frac{a}{\alpha_1} \tag{14.31}$$

thus expressing (14.29) as

$$b = -\alpha_1\beta b + \frac{1}{2}\alpha_2 \left(\frac{a}{\alpha_1}\right)^2 \tag{14.32}$$

That is

$$b(1 + \alpha_1\beta) = \frac{\alpha_2}{2}\left(\frac{a}{\alpha_1}\right)^2 \tag{14.33}$$

$$= \frac{\alpha_2}{2\alpha_1^2} \frac{\alpha_1^2}{(1 + \beta\alpha_1)^2} V_m^2 \tag{14.34}$$

It follows that

$$b = \frac{\alpha_2 V_m^2}{2} \frac{1}{(1 + \beta\alpha_1)^3} \tag{14.35}$$

For a meaningful comparison, we normalize the amplitude of the second harmonic to that of the fundamental:

$$\frac{b}{a} = \frac{\alpha_2 V_m}{2} \frac{1}{\alpha_1} \frac{1}{(1 + \beta\alpha_1)^2} \tag{14.36}$$

Without feedback, on the other hand, such a ratio would be equal to $(\alpha_2 V_m^2/2)/\alpha_1 V_m = \alpha_2 V_m/(2\alpha_1)$. Thus, the relative magnitude of the second harmonic has dropped by a factor of $(1 + \beta\alpha_1)^2$. Negative feedback therefore reduces the relative second harmonic by a factor of $(1 + \beta\alpha_1)^2$ and the gain by $1 + \beta\alpha_1$.

As described in Chapter 8, a feedback circuit employing a feedforward amplifier with a finite gain suffers from gain error. For a feedforward gain of A_0 and a feedback factor of β, the relative gain error is approximately equal to $1/(\beta A_0)$. If the feedforward amplifier exhibits nonlinearity, it is possible to derive a simple relationship between the gain error and the maximum nonlinearity of the overall feedback circuit. As illustrated in Fig. 14.8, we draw two straight lines, one representing the ideal characteristic (with a slope of $1/\beta$) and another passing through the end points of the actual characteristic. We note that with this construction, the nonlinearity, Δy_2, is always smaller than the gain error, Δy_1. This is, of course, true only if the small-signal gain drops monotonically as x goes from 0 to x_{max}, a typical behavior in most analog circuits. Thus, a sufficient condition to ensure that $\Delta y_2 < \epsilon$ is to guarantee that $\Delta y_1 < \epsilon$ by choosing a high open-loop gain for the amplifier.

The above condition is often applied in analog design because it is much easier to predict the open-loop gain than its nonlinearity. Of course, this simplification is obtained at the cost of a pessimistic choice of the amplifier's gain, an issue that becomes more serious as short-channel devices limit the voltage gain that can be achieved.

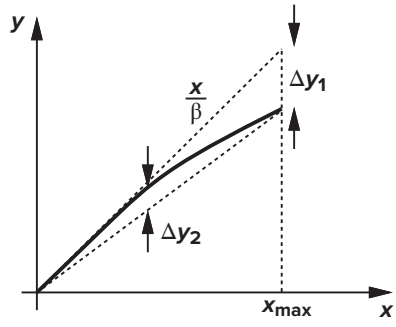


Figure 14.8 Gain error and nonlinearity in a feedback system.

14.1.4 Capacitor Nonlinearity

In switched-capacitor circuits, the voltage dependence of capacitors may introduce substantial distortion. While for a linear capacitor we have $Q = CV$, for a voltage-dependent capacitor we must write $dQ = C\,dV$. Thus, the total charge on a capacitor sustaining a voltage V_1 is

$$Q(V_1) = \int_0^{V_1} C\,dV \tag{14.37}$$

This means that the charge depends on the "history" of the voltage rather than its instantaneous value. In other words, we cannot write $Q(V_1) = CV_1$ even if C is evaluated for a voltage V_1 across the capacitor. To study the effect of capacitor nonlinearity, we express each capacitor as $C = C_0(1 + \alpha_1 V + \alpha_2 V^2 + \cdots)$.

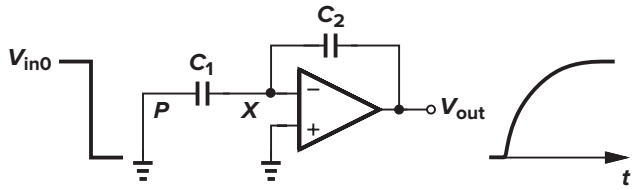


Figure 14.9 Effect of capacitor nonlinearity.

Let us consider the noninverting amplifier of Fig. 13.43(a), repeated in Fig. 14.9, as an example. At the beginning of the amplification mode, C_1 has a voltage equal to V_{in0} and C_2 a voltage of zero. Assuming that $C_1 \approx MC_0(1 + \alpha_1 V)$, where M is the nominal closed-loop gain ($C_1 = MC_2$), we obtain the charge across C_1 as

$$Q_1 = \int_0^{V_{in0}} C_1\,dV \tag{14.38}$$

$$= \int_0^{V_{in0}} MC_0(1 + \alpha_1 V)\,dV \tag{14.39}$$

$$= MC_0 V_{in0} + MC_0 \frac{\alpha_1}{2} V^2 \tag{14.40}$$

Similarly, if $C_2 \approx C_0(1 + \alpha_1 V)$, then the charge on this capacitor at the end of the amplification mode is

$$Q_2 = \int_0^{V_{out}} C_2\,dV \tag{14.41}$$

$$= C_0 V_{out} + C_0 \frac{\alpha_1}{2} V_{out}^2 \tag{14.42}$$

Equating Q_1 and Q_2 and solving for V_{out}, we have

$$V_{out} = \frac{1}{\alpha_1}\left(-1 + \sqrt{1 + M\alpha_1^2 V_{in0}^2 + 2M\alpha_1 V_{in0}}\right) \tag{14.43}$$

The last two terms under the square root are usually much less than unity and, since for $\epsilon \ll 1$, $\sqrt{1+\epsilon} \approx 1 + \epsilon/2 - \epsilon^2/8$, we can write

$$V_{out} \approx MV_{in0} + (1 - M)\frac{M\alpha_1}{2}V_{in0}^2 \tag{14.44}$$

The second term in the above equation represents the nonlinearity resulting from the voltage dependence of the capacitor.

14.1.5 Nonlinearity in Sampling Circuits

Recall from Chapter 13 that the on-resistance of MOS switches in a sampling circuit varies with the input and output levels. For example, the NMOS switch in Fig. 14.10(a) exhibits a rising resistance as V_{in} and V_{out} increase. Similarly, the complementary topology in Fig. 14.10(b) displays an equivalent resistance that varies considerably as V_{in} and V_{out} go from 0 to V_{DD}. In contrast to the monotonic behavior derived in Chapter 13, R_{on} reaches a peak here due to the dependence of the mobility upon the vertical field in the channel. We wish to examine the harmonic distortion observed in the output due to this effect.

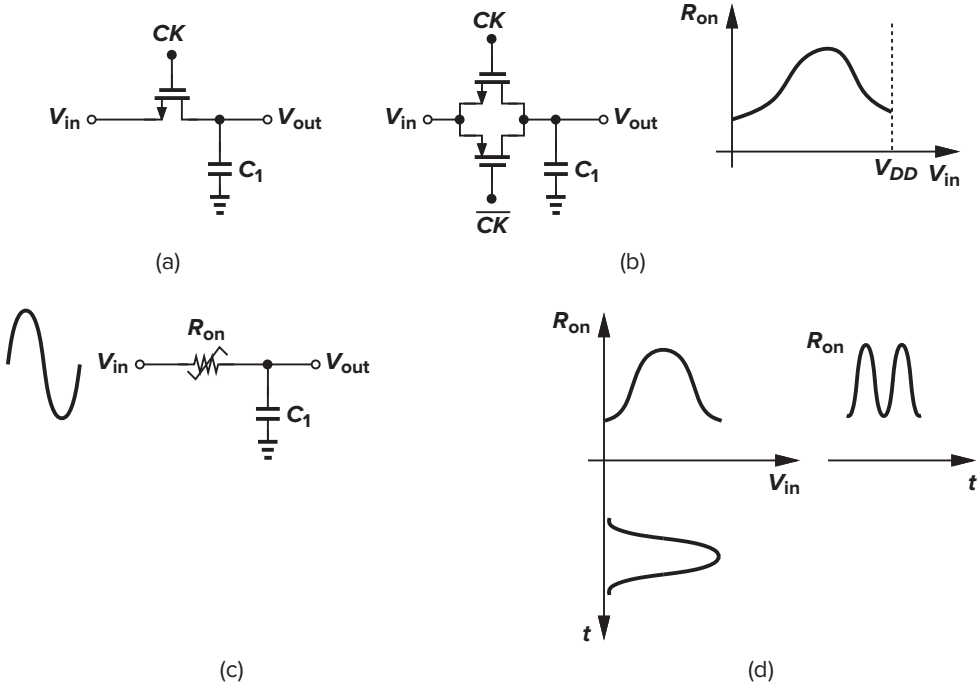


Figure 14.10 (a) Sampling circuit using NMOS switch, (b) sampling circuit using complementary devices, (c) representation of switch on-resistance by a nonlinear resistor, and (d) time-domain behavior.

As shown in Fig. 14.10(c), we apply a large sinusoid to the input, $V_{in} = V_0 \cos \omega_0 t + V_0$, where $V_0 = V_{DD}/2$, and seek the harmonics at the output. How do we analyze this circuit? The nonlinear dependence of R_{on} upon V_{in} or V_{out} presents a formidable challenge. Let us first assume that the resistance

is linear and write the output as

$$V_{out}(t) = \frac{V_0}{\sqrt{R_{on}^2 C_1^2 \omega_0^2 + 1}} \cos[\omega_0 t - \tan^{-1}(R_{on} C_1 \omega_0)] + V_0 \tag{14.45}$$

In practice, the bandwidth must be large enough to negligibly attenuate the signal, i.e., $R_{on} C_1 \omega_0 \ll 1$, leading to

$$V_{out}(t) \approx V_0 \cos(\omega_0 t - R_{on} C_1 \omega_0) + V_0 \tag{14.46}$$

We now assume that this expression also holds for the nonlinear circuit if R_{on} is represented properly. It is interesting to note that R_{on}, and hence the *phase shift* from the input to the output, vary as V_{in} and V_{out} go up and down, thus creating distortion.

A key observation that simplifies our analysis is that, with a periodic input, R_{on} also varies *periodically* and can therefore be approximated by a Fourier series:

$$R_{on}(t) = R_0 + R_1 \cos \omega_0 t + R_2 \cos(2\omega_0 t) + \cdots \tag{14.47}$$

If we assume a roughly symmetric behavior for R_{on} in Fig. 14.10(b), we observe the time-domain behavior depicted in Fig. 14.10(d), where R_{on} varies at a rate equal to twice the input frequency. In this special case, $R_1 \approx 0$, but we continue with the general case. Replacing for R_{on} in Eq. (14.46), we have

$$V_{out}(t) \approx V_0 \cos[\omega_0 t - R_0 C_1 \omega_0 - R_1 C_1 \omega_0 \cos \omega_0 t - R_2 C_1 \omega_0 \cos(2\omega_0 t) - \cdots] + V_0 \tag{14.48}$$

If the cosine terms in the argument have amplitudes much less than 1 rad,

$$\begin{aligned} V_{out}(t) \approx\ & V_0 \cos(\omega_0 t - R_0 C_1 \omega_0) + \\ & [R_1 C_1 \omega_0 \cos \omega_0 t + R_2 C_1 \omega_0 \cos(2\omega_0 t) + \cdots] V_0 \sin(\omega_0 t - R_0 C_1 \omega_0) + V_0 \end{aligned}$$

We observe that the products $\cos \omega_0 t \sin(\omega_0 t - R_0 C_1 \omega_0)$, $\cos(2\omega_0 t) \sin(\omega_0 t - R_0 C_1 \omega_0)$, etc., give rise to harmonics. For example, the first two products respectively translate to a second harmonic and a third harmonic having peak amplitudes of $V_0 R_1 C_1 \omega_0 / 2$ and $V_0 R_2 C_1 \omega_0 / 2$, respectively. If we retain only these two harmonics, then

$$\text{THD} = \frac{R_1^2 + R_2^2}{4} C_1^2 \omega_0^2 \tag{14.49}$$

In a differential sampling switch, the even-order harmonics are suppressed.

14.1.6 Linearization Techniques

While amplifiers using "global" feedback (e.g., the switched-capacitor topologies of Chapter 13) can achieve a high linearity, stability and settling issues of feedback circuits limit their usage in high-speed applications. For this reason, many other techniques have been invented to linearize amplifiers with less compromise in speed.

The principle behind linearization is to reduce the dependence of the circuit's gain upon the input level. This usually translates into making the gain relatively independent of the transistor bias currents.

The simplest linearization method is source degeneration by means of a linear resistor. As shown in Fig. 14.11 for a common-source stage and revealed by the observations in the previous section, degeneration reduces the signal swing applied between the gate and the source of the transistor, thereby

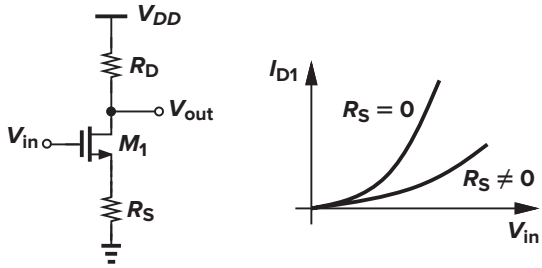


Figure 14.11 Common-source stage with resistive degeneration.

making the input/output characteristic more linear. From another point of view, neglecting body effect, we can write the overall transconductance of the stage as

$$G_m = \frac{g_m}{1 + g_m R_S} \tag{14.50}$$

which for large $g_m R_S$ approaches $1/R_S$, an input-independent value.

Note that the amount of linearization depends on $g_m R_S$ rather than on R_S alone. With a relatively constant G_m, the voltage gain, $G_m R_D$, is also relatively independent of the input and the amplifier is linearized.

▶ Example 14.2

A common-source stage biased at a current I_1 experiences an input voltage swing that varies the drain current from $0.75I_1$ to $1.25I_1$. Calculate the variation of the small-signal voltage gain (a) with no degeneration and (b) with degeneration such that $g_m R_S = 2$, where g_m denotes the transconductance at $I_D = I_1$.

Solution

Assuming square-law behavior, we have $g_m \propto \sqrt{I_D}$. For the case of no degeneration,

$$\frac{g_{m,high}}{g_{m,low}} = \sqrt{\frac{1.25}{0.75}} \tag{14.51}$$

With $g_m R_S = 2$,

$$\frac{G_{m,high}}{G_{m,low}} = \frac{\dfrac{\sqrt{1.25}g_m}{1 + \sqrt{1.25}g_m R_S}}{\dfrac{\sqrt{0.75}g_m}{1 + \sqrt{0.75}g_m R_S}} \tag{14.52}$$

$$= \sqrt{\frac{1.25}{0.75}} \cdot \frac{1 + 2\sqrt{0.75}}{1 + 2\sqrt{1.25}} \tag{14.53}$$

$$= 0.84\sqrt{\frac{1.25}{0.75}} \tag{14.54}$$

Thus, degeneration decreases the variation of the small-signal gain by approximately 16% in this case. ◀

Resistive degeneration presents trade-offs among linearity, noise, power dissipation, and gain. For large input voltage swings (e.g., 0.5 V_{pp}), it may be difficult to achieve even a voltage gain of 2 in a common-source stage if the nonlinearity is to remain below 1%.

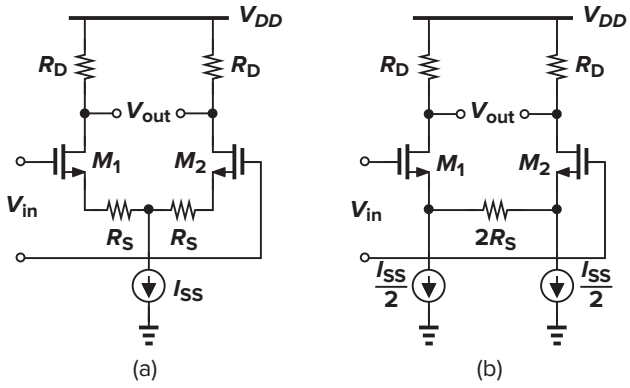


Figure 14.12 Source degeneration applied to a differential pair.

A differential pair can be degenerated as shown in Figs. 14.12(a) and (b). In Fig. 14.12(a), I_{SS} flows through the degeneration resistors, thereby consuming a voltage headroom of $I_{SS}R_S/2$, an important issue if a high level of degeneration is required. The circuit of Fig. 14.12(b), on the other hand, does not involve this issue, but it suffers from a slightly higher noise (and offset voltage) because the two tail current sources introduce some differential error. The reader can prove that if the output noise current of each current source is equal to $\overline{I_n^2}$, then the input-referred noise voltage of the circuit of Fig. 14.12(b) is higher than that of Fig. 14.12(a) by $2\overline{I_n^2}R_S^2$.

As depicted in Fig. 14.13, the resistor can be replaced by a MOSFET operating in the deep triode region. However, for large input swings, M_3 may not remain in the deep triode region, thereby experiencing substantial change in its on-resistance. Furthermore, V_b must track the input common-mode level so that R_{on3} can be defined accurately.

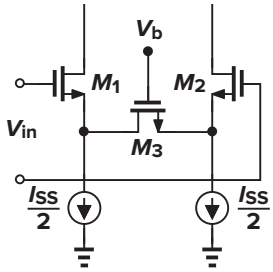


Figure 14.13 Differential pair degenerated by a MOSFET operating in the deep triode region.

A more practical variant of the above idea is illustrated in Fig. 14.14 [1]. Here, M_3 and M_4 are in the deep triode region if $V_{in} = 0$. As the gate voltage of M_1 becomes more positive than the gate voltage of M_2, transistor M_3 stays in the triode region because $V_{D3} = V_{G3} - V_{GS1}$, whereas M_4 eventually enters

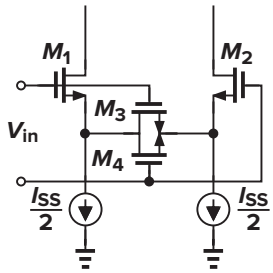


Figure 14.14 Differential pair degenerated by two MOSFETs operating in the triode region.

the saturation region because its drain voltage rises and its gate and source voltages fall. Thus, the circuit remains relatively linear even if one degeneration device goes into saturation. For the widest linear region, [1] suggests that $(W/L)_{1,2} \approx 7(W/L)_{3,4}$.

▶ Example 14.3

Using the offset mechanism introduced in Fig. 4.19, devise another linearization technique.

Solution

From Example 4.6, we know that the mismatch between the transistors' widths shifts the characteristics horizontally. Let us create a negative shift and a positive shift (equal in magnitude) in two differential pairs [Fig. 14.15(a)], observing that their G_m plots are offset by equal and opposite amounts. We now add the output currents by simply shorting the corresponding drains as shown in Fig. 14.15(b). The G_m plots also add (why?), yielding a result that is relatively constant for a *wider* range of $V_{in1} - V_{in2}$ and represents a more linear circuit. The ratio of 2 between the two widths is merely an example to illustrate the technique and may need to be modified for optimum linearity.

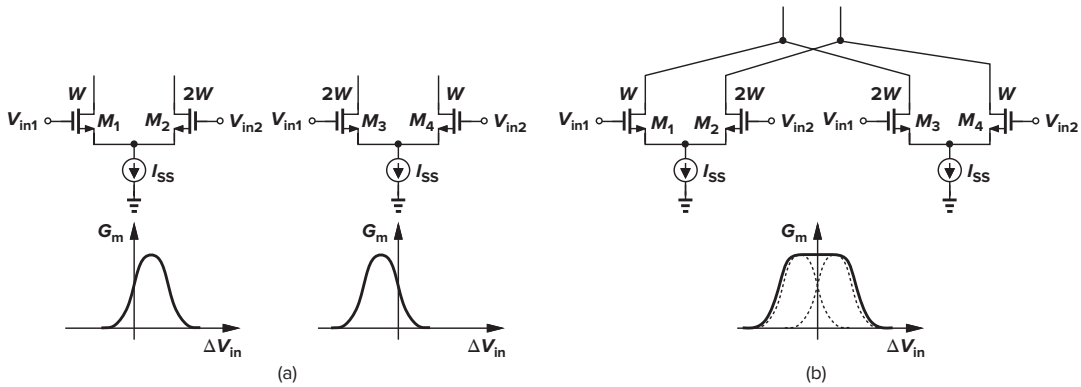


Figure 14.15

A linearization technique that avoids the use of resistors is based on the observation that a MOSFET operating in the triode region can provide a linear I_D/V_{GS} characteristic if its drain-source voltage is held constant: $I_D = (1/2)\mu C_{ox}(W/L)[2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2]$. Illustrated in Fig. 14.16, the technique employs amplifiers A_1 and A_2 along with cascode devices M_3 and M_4 to force V_X and V_Y to be equal to V_b for varying input levels.

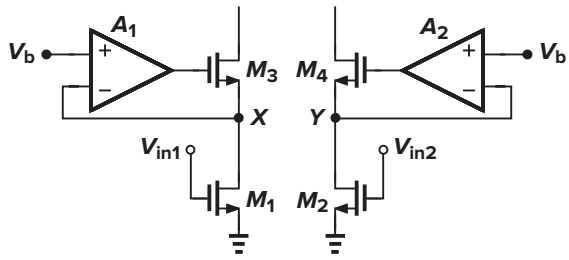


Figure 14.16 Differential pair using input devices operating in the triode region.

This circuit suffers from several drawbacks. First, the transconductance of M_1 and M_2, equal to $\mu_n C_{ox}(W/L)V_{DS}$, is relatively small because V_{DS} must be low enough to ensure that each input transistor remains in the triode region. Second, the input common-mode level must be tightly controlled, and it must track V_b so as to define I_{D1} and I_{D2}. Third, M_3, M_4, and the two auxiliary amplifiers contribute substantial noise to the output.

Another approach to linearizing voltage amplifiers is to perform "post correction." Illustrated in Fig. 14.17, the idea is to view the amplifier as a voltage-to-current (V/I) converter followed by a current-to-voltage (I/V) converter. If the V/I converter can be described as $I_{out} = f(V_{in})$ and the I/V converter as $V_{out} = f^{-1}(I_{in})$, then V_{out} is a linear function of V_{in}. That is, the second stage corrects the nonlinearity introduced by the first stage. As an example, recall from Chapter 4 that for the circuit shown in Fig. 14.18(a), we have

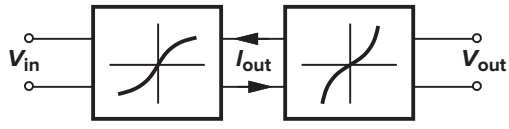


Figure 14.17 Voltage amplifier viewed as a cascade of two nonlinear stages.

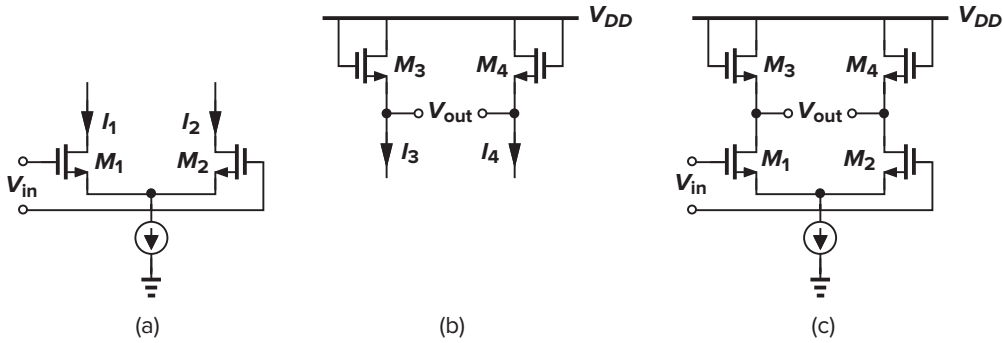


Figure 14.18 (a) Differential pair with nonlinear I/V characteristic; (b) diode-connected devices with nonlinear V/I characteristic; (c) circuit having linear input/output characteristic.

$$V_{in1} - V_{in2} = V_{GS1} - V_{GS2} \tag{14.55}$$

$$= \sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2}}} - \sqrt{\frac{2I_{D2}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2}}} \tag{14.56}$$

We also note that for the circuit shown in Fig. 14.18(b),

$$V_{out} = V_{GS3} - V_{GS4} \tag{14.57}$$

$$= \sqrt{\frac{2I_3}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{3,4}}} - \sqrt{\frac{2I_4}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{3,4}}} \tag{14.58}$$

where channel-length modulation and body effect are neglected. It follows that for the circuit shown in Fig. 14.18(c),

$$V_{out} = \sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{3,4}}} - \sqrt{\frac{2I_{D2}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{3,4}}} \tag{14.59}$$

$$= \frac{1}{\sqrt{\left(\frac{W}{L}\right)_{3,4}}}(V_{in1} - V_{in2})\, sqrt\left(\frac{W}{L}\right)_{1,2} \tag{14.60}$$

Thus, as derived in Chapter 4, the voltage gain is equal to

$$A_v = \sqrt{\frac{\left(\frac{W}{L}\right)_{1,2}}{\left(\frac{W}{L}\right)_{3,4}}} \tag{14.61}$$

a quantity independent of the bias currents of the transistors.

In practice, body effect and other nonidealities in short-channel devices give rise to nonlinearity in this circuit. Furthermore, as the differential input level increases, driving M_1 or M_2 into the subthreshold region, Eqs. (14.56) and (14.58) no longer hold and the gain drops sharply.

It is possible to add local feedback to a degenerated differential pair to linearize it further. Illustrated in Fig. 14.19(a), the idea is to sense the output voltage of the differential pair by means of M_3 and M_4 and return a proportional current to the *sources* of M_1 and M_2. The reader can readily prove that the feedback is negative. We assume that the circuit is symmetric and $I_1 = \cdots = I_4$.

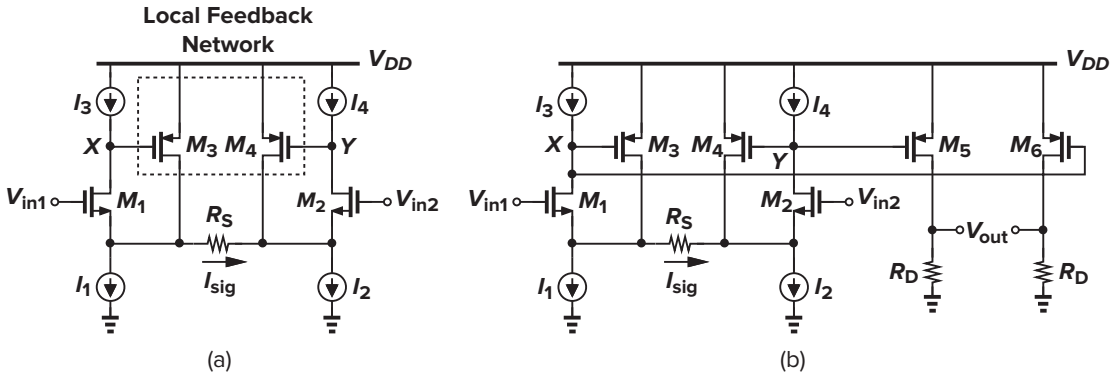


Figure 14.19 (a) Differential pair with local feedback, and (b) use of (a) in voltage amplification.

If channel-length modulation and body effect are neglected, we observe that $I_{D1} = I_3$ and $I_{D2} = I_4$ *regardless* of the input signal. Thus, the input transistors maintain a constant V_{GS} as $V_{in} = V_{in1} - V_{in2}$ varies. Moreover, since $I_1 = I_3 = I_{D1}$ and $I_2 = I_4 = I_{D2}$, the current flowing through R_S must be provided by only M_3 and M_4. Denoting this current by I_{sig}, we have

$$V_{in} = V_{GS1} + I_{sig}R_S - V_{GS2} \tag{14.62}$$

$$= I_{sig}R_S \tag{14.63}$$

Interestingly, the currents produced by M_3 and M_4 are linearly proportional to V_{in} because the feedback from their drains to their sources guarantees a constant V_{GS}. Note, however, that $V_X - V_Y$ is not.

The reader may wonder where this topology's output is! As depicted in Fig. 14.19(b), we copy the PMOS currents onto M_5 and M_6 and allow the results to flow through (linear) resistors. Since I_{D3} and I_{D4} are equal and opposite, it follows from (14.63) that

$$V_{out} = \frac{2R_D}{R_S} V_{in} \tag{14.64}$$

where the PMOS devices are assumed identical. The circuit excluding the R_D's operates as a linear voltage-to-current converter ("transconductor").

The above topology entails two issues. First, the large number of devices in the signal path produces substantial noise. In addition to M_1–M_4, the top and bottom current sources also contribute differential noise. Second, due to the dependence of r_O upon V_{DS} in short-channel devices (Chapter 17), the output stage introduces some nonlinearity.

14.2 ■ Mismatch

Our study of differential amplifiers in the previous chapters has mostly assumed that the circuits are perfectly symmetric, i.e., the two sides exhibit identical properties and bias currents. In reality, however, nominally-identical devices suffer from a finite mismatch due to uncertainties in each step of the manufacturing process. For example, as illustrated in Fig. 14.20, the gate dimensions of MOSFETs suffer from random, microscopic variations, introducing mismatches between the equivalent lengths and widths of two transistors that are identically laid out. Also, MOS devices exhibit threshold voltage mismatch because, from Eq. (2.1), V_{TH} is a function of the doping levels in the channel and the gate, and these levels vary randomly from one device to another.

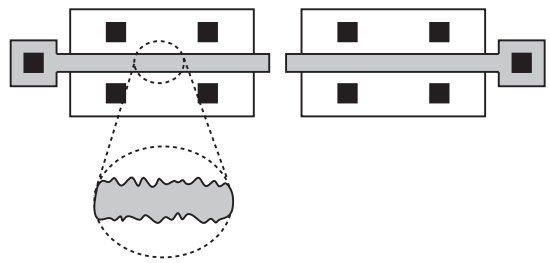


Figure 14.20 Random mismatches due to microscopic variations in device dimensions.

Study of mismatch consists of two steps: (1) identify and formulate the mechanisms that lead to mismatch between devices; and (2) analyze the effect of device mismatches upon the performance of circuits. Unfortunately, the first step is quite complex and heavily dependent on the fabrication technology and the layout, often requiring actual measurements of mismatches. For example, the achievable mismatch between capacitors is typically quoted to be 0.1%, but this value is not derived from any fundamental quantities. Layout techniques for minimum mismatch are described in Chapter 19.

Expressing the characteristics of a MOSFET in saturation as $I_D = (1/2)\mu C_{ox}(W/L)(V_{GS} - V_{TH})^2$, we observe that mismatches between μ, C_{ox}, W, L, and V_{TH} result in mismatches between drain currents (for a given V_{GS}) or gate-source voltages (for a given drain current) of two nominally-identical transistors. Intuitively, we expect that as W and L increase, their relative mismatches, $\Delta W/W$ and $\Delta L/L$, respectively, decrease, i.e., larger devices exhibit smaller mismatches. A more important observation is that all of the mismatches decrease as the *area* of the transistor, WL, increases. For example, increasing W

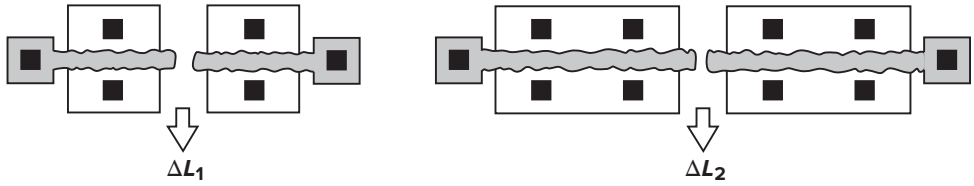


Figure 14.21 Reduction of length mismatch as a result of increasing the width.

reduces both $\Delta W/W$ *and* $\Delta L/L$. This is because as WL increases, random variations experience greater "averaging," thereby falling in magnitude. For the case depicted in Fig. 14.21, $\Delta L_2 < \Delta L_1$ because, if the device is viewed as many small parallel transistors (Fig. 14.22), each having a width of W_0, then we can write the equivalent length as $L_{eq} \approx (L_1 + L_2 + \cdots + L_n)/n$. The overall variation is therefore given by

$$\Delta L_{eq} \approx \left(\Delta L_1^2 + \Delta L_2^2 + \cdots + \Delta L_n^2\right)^{1/2}/n \tag{14.65}$$

$$= \frac{\left(n\Delta L_0^2\right)^{1/2}}{n} \tag{14.66}$$

$$= \frac{\Delta L_0}{\sqrt{n}} \tag{14.67}$$

where ΔL_0 is the statistical variation of the length for a transistor with width W_0. Equation (14.67) reveals that for a given W_0, as n increases, the variation of L_{eq} decreases.

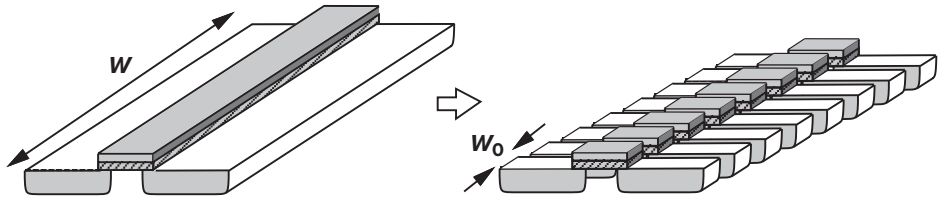


Figure 14.22 Wide MOSFET viewed as a parallel combination of narrow devices.

The above result can be extended to other device parameters as well. For example, we postulate that μC_{ox} and V_{TH} suffer from less mismatch if the device area increases. Illustrated in Fig. 14.23, the reason is that a large transistor can be decomposed into a series and parallel combination of small unit transistors with dimensions W_0 and L_0, each exhibiting $(\mu C_{ox})_j$ and V_{THj}. For given W_0 and L_0, as the number of unit transistors increases, μC_{ox} and V_{TH} experience greater averaging, leading to smaller mismatch between two large transistors.

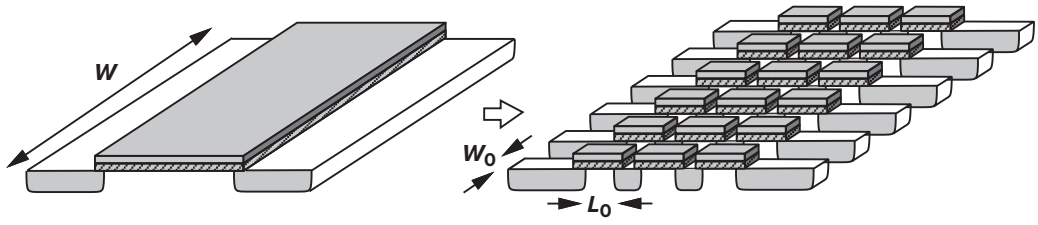


Figure 14.23 Large MOSFET viewed as a combination of small devices.

The foregoing qualitative observations have been verified mathematically and experimentally [2, 3]. Here, we state without proof that

$$\Delta V_{TH} = \frac{A_{VTH}}{\sqrt{WL}} \tag{14.68}$$

$$\Delta\left(\mu C_{OX}\frac{W}{L}\right) = \frac{A_K}{\sqrt{WL}} \tag{14.69}$$

where A_{VTH} and A_K are proportionality factors and obtained from measurements.

▶ **Example 14.4**

A differential pair incorporates transistors having a length of 40 nm. If $A_{VTH} = 4\ \text{mV}\cdot\mu\text{m}$ for 40-nm technology, what is the minimum device width that guarantees $\Delta V_{TH} \leq 2$ mV?

Solution

We write

$$W = \frac{A_{VTH}^2}{L\Delta V_{TH}^2} \tag{14.70}$$

$$= 100\ \mu\text{m} \tag{14.71}$$

We observe the very large W/L necessary for low offsets in nanometer technologies. ◀

Since the channel capacitance is proportional to WLC_{ox}, we note that ΔV_{TH} and the channel capacitance bear a trade-off.

14.2.1 Effect of Mismatch

We now study the effect of device mismatch upon the performance of circuits. Mismatches lead to three significant phenomena: dc offsets, finite even-order distortion, and lower common-mode rejection. The last phenomenon was studied in Chapter 4.

DC Offsets Consider the differential pair shown in Fig. 14.24(a). With $V_{in} = 0$ and perfect symmetry, $V_{out} = 0$, but in the presence of mismatches, $V_{out} \neq 0$. We say that the circuit suffers from a dc "offset" equal to the observed value of V_{out} when V_{in} is set to zero. In practice, it is more meaningful to specify the input-referred offset voltage, defined as the input level that forces the output voltage to go to zero

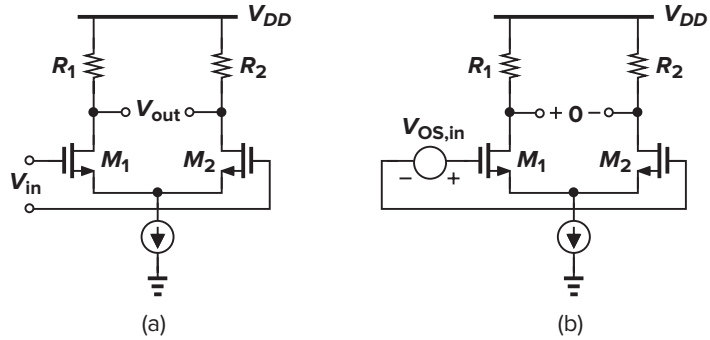


Figure 14.24 (a) Differential pair with offset measured at the output; (b) circuit of (a) with its offset referred to the input.

[Fig. 14.24(b)]. Note that $|V_{OS,in}| = |V_{OS,out}|/A_v$. As with random noise, the polarity of random offsets is unimportant.

How does offset limit the performance? Suppose the differential pair of Fig. 14.24 is to amplify a small input voltage. Then, as depicted in Fig. 14.25, the output contains amplified replicas of both the signal and the offset. In a cascade of direct-coupled amplifiers, the dc offset may experience so much gain that it drives the latter stages into nonlinear operation.

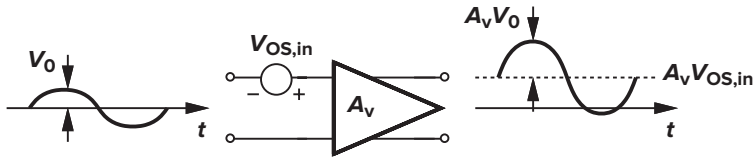


Figure 14.25 Effect of offset in an amplifier.

A more important effect of offset is the limitation on the precision with which signals can be measured. For example, if an amplifier is used to determine whether the input signal is greater or less than a reference, V_{REF} (Fig. 14.26), then the input-referred offset imposes a lower bound on the minimum $V_{in} - V_{REF}$ that can be detected reliably.

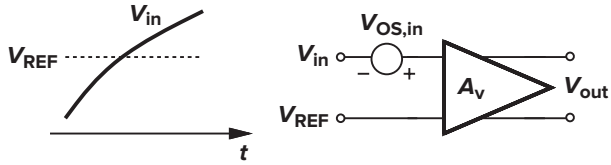


Figure 14.26 Accuracy limitation of an amplifier due to offset.

Let us now calculate the offset voltage of a differential pair, assuming that both the input transistors and the load resistors suffer from mismatch. As illustrated in Fig. 14.24(b), our objective is to find the value of $V_{OS,in}$ such that $V_{out} = 0$. The device mismatches are incorporated as $V_{TH1} = V_{TH}$, $V_{TH2} = V_{TH} + \Delta V_{TH}$; $(W/L)_1 = W/L$, $(W/L)_2 = W/L + \Delta(W/L)$; $R_1 = R_D$, $R_2 = R_D + \Delta R$. For simplicity, $\lambda = \gamma = 0$, and mismatches in $\mu_n C_{ox}$ are neglected. For $V_{out} = 0$, we must have $I_{D1}R_1 = I_{D2}R_2$, concluding that I_{D1} cannot be equal to I_{D2}. Thus, we assume that $I_{D1} = I_D$, $I_{D2} = I_D + \Delta I_D$.

Since $V_{OS,in} = V_{GS1} - V_{GS2}$, we have

$$V_{OS,in} = \sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_1}} + V_{TH1} - \sqrt{\frac{2I_{D2}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_2}} - V_{TH2} \tag{14.72}$$

$$= \sqrt{\frac{2}{\mu_n C_{ox}}}\left[\sqrt{\frac{I_D}{\frac{W}{L}}} - \sqrt{\frac{I_D + \Delta I_D}{\frac{W}{L} + \Delta\left(\frac{W}{L}\right)}}\right] - \Delta V_{TH} \tag{14.73}$$

$$= \sqrt{\frac{2}{\mu_n C_{ox}}}\sqrt{\frac{I_D}{W/L}}\left[1 - \sqrt{\frac{1 + \dfrac{\Delta I_D}{I_D}}{1 + \Delta\left(\dfrac{W}{L}\right)\Big/\left(\dfrac{W}{L}\right)}}\right] - \Delta V_{TH} \tag{14.74}$$

Assuming that $\Delta I_D/I_D$ and $\Delta(W/L)/(W/L) \ll 1$, and noting that for $\epsilon \ll 1$ we can write $\sqrt{1+\epsilon} \approx 1+\epsilon/2$ and $(\sqrt{1+\epsilon})^{-1} \approx 1-\epsilon/2$, we reduce (14.74) to

$$V_{OS,in} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}\left\{1-\left(1+\frac{\Delta I_D}{2I_D}\right)\left[1-\frac{\Delta(W/L)}{2(W/L)}\right]\right\} - \Delta V_{TH} \tag{14.75}$$

$$= \sqrt{\frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}\left[\frac{-\Delta I_D}{2I_D}+\frac{\Delta(W/L)}{2(W/L)}\right] - \Delta V_{TH} \tag{14.76}$$

where the product of two small quantities is neglected. Recall that $I_{D1}R_1 = I_{D2}R_2$, and hence $I_D R_D = (I_D+\Delta I_D)(R_D+\Delta R_D) \approx I_D R_D + R_D\Delta I_D + I_D \Delta R_D$. Consequently, $\Delta I_D/I_D \approx -\Delta R_D/R_D$, and

$$V_{OS,in} = \frac{1}{2}\sqrt{\frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}\left[\frac{\Delta R_D}{R_D}+\frac{\Delta(W/L)}{(W/L)}\right] - \Delta V_{TH} \tag{14.77}$$

We also recognize that the square-root quantity is approximately equal to the equilibrium overdrive voltage of each transistor, $V_{GS}-V_{TH}$, and

$$V_{OS,in} = \frac{V_{GS}-V_{TH}}{2}\left[\frac{\Delta R_D}{R_D}+\frac{\Delta(W/L)}{(W/L)}\right] - \Delta V_{TH} \tag{14.78}$$

Equation (14.78) is an important result, revealing the dependence of $V_{OS,in}$ on device mismatches and bias conditions. We note that (1) the contribution of load resistor mismatch and transistor dimension mismatch *increases* with the equilibrium overdrive, and (2) the threshold voltage mismatch is directly referred to the input. Thus, it is desirable to minimize $V_{GS}-V_{TH}$ by lowering the tail current or increasing the transistor widths. In reality, since mismatches are independent statistical variables, we express (14.78) as[2]

$$V_{OS,in}^2 = \left(\frac{V_{GS}-V_{TH}}{2}\right)^2\left\{\left(\frac{\Delta R_D}{R_D}\right)^2+\left[\frac{\Delta(W/L)}{(W/L)}\right]^2\right\} + \Delta V_{TH}^2 \tag{14.79}$$

where squared quantities represent standard deviations.

To gain more insight into the effect of offset, let us establish an analogy between offset and *noise*. If the two inputs of a differential pair are shorted, the output voltage exhibits a finite noise, that is, a voltage that varies with time. We may therefore say that the offset voltage of a differential pair resembles a very low-frequency noise component, varying so slowly that it appears constant in our measurements. Viewed as such, offsets can be incorporated as noise sources, allowing us to utilize the analysis techniques developed in Chapter 7. To this end, we represent the offset of two nominally-identical transistors by a voltage source equal to (14.79) in series with the gate of one of the transistors.

[2] As mentioned earlier, ΔV_{TH} does depend on W, an effect that can be added as a cross-correlation term. We neglect this term here for simplicity.

▶ Example 14.5

Calculate the input-referred offset voltage of the circuit shown in Fig. 14.27(a). Assume all of the transistors operate in saturation.

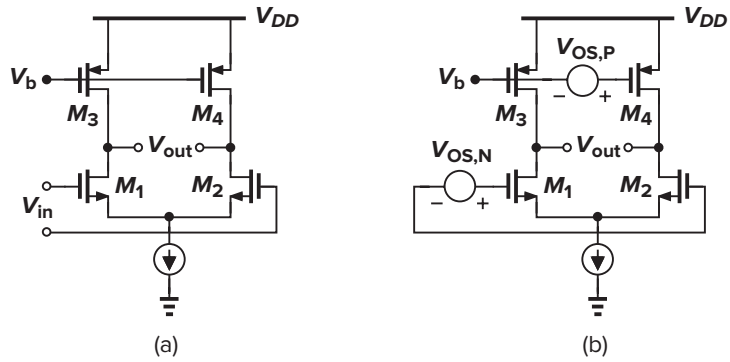


Figure 14.27

Solution

We insert the offsets of the NMOS and PMOS pairs as in Fig. 14.27(b). To obtain $I_{D1} = I_{D2}$ and $I_{D3} = I_{D4}$, we have from (14.78)

$$V_{OS,N} = \frac{(V_{GS} - V_{TH})_N}{2}\left[\frac{\Delta(W/L)}{W/L}\right]_N + \Delta V_{TH,N} \tag{14.80}$$

$$V_{OS,P} = \frac{|V_{GS} - V_{TH}|_P}{2}\left[\frac{\Delta(W/L)}{W/L}\right]_P + \Delta V_{TH,P} \tag{14.81}$$

From the noise analysis in Chapter 7, $V_{OS,P}$ is amplified by a gain of $g_{mP}(r_{ON}||r_{OP})$ and divided by $g_{mN}(r_{ON}||r_{OP})$ when referred to the main input. As a result,

$$V_{OS,in} = \left\{\frac{|V_{GS} - V_{TH}|_P}{2}\left[\frac{\Delta(W/L)}{W/L}\right]_P + \Delta V_{TH,P}\right\}\frac{g_{mP}}{g_{mN}} + \frac{(V_{GS} - V_{TH})_N}{2}\left[\frac{\Delta(W/L)}{W/L}\right]_N + \Delta V_{TH,N} \tag{14.82}$$

In practice, we add the "power" of these terms, as exemplified by (14.79). Note that, as with noise, the contribution of the offset of the PMOS pair is proportional to g_{mP}/g_{mN}.

◀

The foregoing example can be better understood if we study the offset behavior of current sources. Consider the nominally-identical current sources M_1 and M_2 in Fig. 14.28. Neglecting channel-length modulation, we determine the total mismatch between I_{D1} and I_{D2} by calculating the total differential.

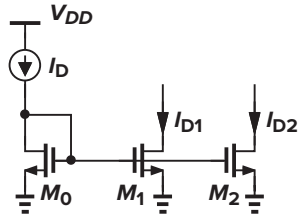


Figure 14.28 Mismatch between two current sources.

Recall from calculus that if $y = f(x_1, x_2, \ldots)$, then the total differential is given by

$$\Delta y = \frac{\partial f}{\partial x_1}\Delta x_1 + \frac{\partial f}{\partial x_2}\Delta x_2 + \cdots \tag{14.83}$$

Equation (14.83) simply means that each mismatch component Δx_j is weighted by the corresponding sensitivity $\partial f/\partial x_j$ as it contributes to the total mismatch. Since $I_D = (1/2)\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})^2$, we have

$$\Delta I_D = \frac{\partial I_D}{\partial (W/L)}\Delta\left(\frac{W}{L}\right) + \frac{\partial I_D}{\partial (V_{GS} - V_{TH})}\Delta(V_{GS} - V_{TH}) \tag{14.84}$$

where mismatches in $\mu_n C_{ox}$ are neglected. It follows that

$$\Delta I_D = \frac{1}{2}\mu_n C_{ox}(V_{GS} - V_{TH})^2\Delta\left(\frac{W}{L}\right) - \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})\Delta V_{TH} \tag{14.85}$$

Unlike the input-referred offset *voltage*, current mismatch is usually normalized to the average value to allow a meaningful comparison:

$$\frac{\Delta I_D}{I_D} = \frac{\Delta(W/L)}{W/L} - 2\frac{\Delta V_{TH}}{V_{GS} - V_{TH}} \tag{14.86}$$

This result suggests that, to minimize current mismatch, the overdrive voltage must be *maximized*, a trend opposite to that in (14.78). This is because as $V_{GS} - V_{TH}$ increases, threshold mismatch has a lesser effect on the device currents.

The dependence of offset voltage and current mismatches upon the overdrive voltage is similar to our observations in Chapter 7 for corresponding noise quantities. For a given current, the input noise voltage of a differential pair increases as the overdrive increases because $g_m = 2I_D/(V_{GS} - V_{TH})$. Also, the output noise current of current sources is proportional to g_m and hence proportional to $V_{GS} - V_{TH}$.

Even-Order Distortion Our study of nonlinearity in Sec. 14.1 implies that, by virtue of odd symmetry, differential circuits are free from even-order distortion. In reality, however, mismatches degrade the symmetry, thereby introducing a finite even-order nonlinearity.

Analysis of the even-order distortion in the presence of mismatches is generally quite complex, often necessitating simulations. Here, we consider a simple case to gain some insight. Suppose the two signal paths in a differential circuit are represented by $y_1 \approx \alpha_1 x_1 + \alpha_2 x_1^2 + \alpha_3 x_1^3$ and $y_2 \approx \beta_1 x_2 + \beta_2 x_2^2 + \beta_3 x_2^3$ (Fig. 14.29). The differential output is given by

$$y_1 - y_2 = (\alpha_1 x_1 - \beta_2 x_2) + \left(\alpha_2 x_1^2 - \beta_2 x_2^2\right) + \left(\alpha_3 x_1^3 - \beta_3 x_2^3\right) \tag{14.87}$$

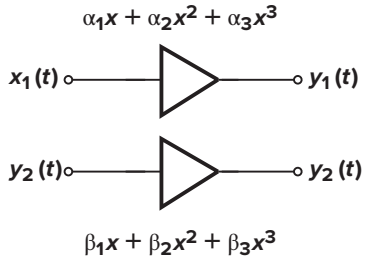


Figure 14.29 Effect of mismatch on second-order distortion.

which, for $x_1 = -x_2$, reduces to

$$y_1 - y_2 = (\alpha_1 + \beta_1)x_1 + (\alpha_2 - \beta_2)x_1^2 + (\alpha_3 + \beta_3)x_1^3 \tag{14.88}$$

If $x_1(t) = A\cos\omega t$, then the second harmonic has an amplitude equal to $(\alpha_2 - \beta_2)A^2/2$, i.e., proportional to the mismatch between the second-order coefficients of the input/output characteristic.

We should also mention that since at high frequencies, signals experience considerable phase shift, even-order distortion may arise from *phase* mismatch. This point is considered in Problem 14.14.1.

In circuits dissipating a high power, thermal gradients across the chip may create asymmetries. For example, if one transistor of a differential pair is closer to a high-power output stage than the other transistor, then mismatches arise between the threshold voltages and the mobilities of the two transistors.

14.2.2 Offset Cancellation Techniques

As mentioned above, the threshold voltage mismatch of MOSFETS trades with the channel capacitance. For example, a threshold mismatch of 1 mV translates to roughly 300 fF of channel capacitance for each transistor in a 0.6-μm technology. If many differential pairs are connected in parallel (e.g., in an A/D converter), the input capacitance becomes prohibitively large, severely degrading the speed and/or demanding high power dissipation in the preceding stage. Another difficulty is that mechanical stress may increase the offset voltages after a circuit is packaged. For these reasons, many high-precision systems require electronic cancellation of the offsets. As explained below, offset cancellation can also reduce $1/f$ noise of amplifiers considerably.

As our first step toward understanding the principle of offset cancellation, let us consider the circuit of Fig. 14.30(a), where a differential amplifier having an input-referred offset voltage V_{OS} is followed by two series capacitors. Now suppose, as shown in Fig. 14.30(b), the inputs are shorted together, driving the amplifier output to $V_{out} = A_v V_{OS}$. Furthermore, assume that during this period, nodes X and Y are shorted together as well. We note that when all of the node voltages are settled and $A_v V_{OS}$ is stored across C_1 and C_2, a zero differential input results in a zero difference between V_X and V_Y. Thus, after S_1 and S_2 turn off, the circuit consisting of the amplifier and C_1 and C_2 exhibits a zero offset voltage, amplifying only *changes* in the differential input voltage. In practice, the inputs and outputs must be shorted to proper common-mode voltages [Fig. 14.30(c)].

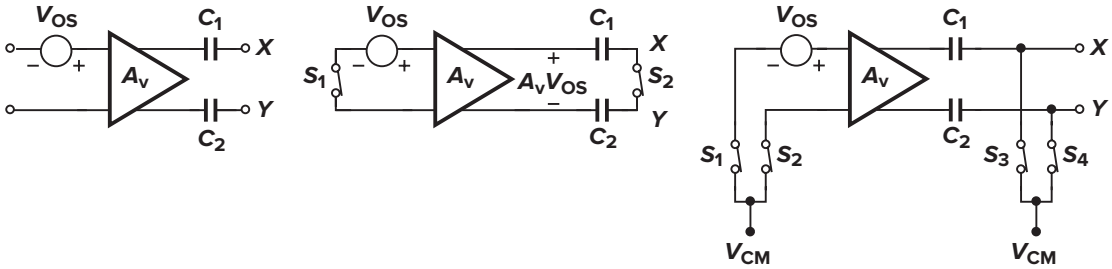


Figure 14.30 (a) Simple amplifier with capacitive coupling at the output; (b) circuit of (a) with its inputs and outputs shorted; (c) proper setting of the common-mode level during offset cancellation.

In summary, this type of offset cancellation "measures" the offset by setting the differential input to zero and stores the result on capacitors in series with the output. The circuit therefore requires a dedicated offset cancellation period, during which the actual input is disabled. Figure 14.31 depicts the final topology, where CK denotes the offset cancellation command. Called "output offset storage," this

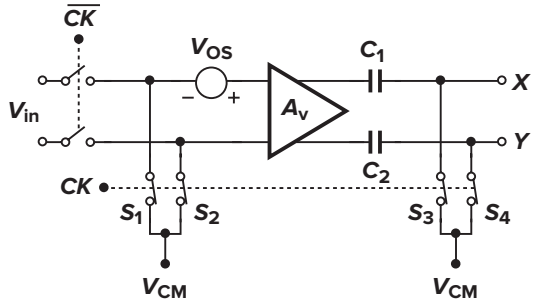


Figure 14.31 Control of amplification and offset cancellation modes by a clock.

technique reduces the overall offset to zero if S_3-S_4 exhibit no charge injection mismatch. Note, however, that if A_v is large, $A_v V_{OS}$ may "saturate" the amplifier output. For this reason, A_v is typically chosen to be less than roughly 10.

In applications where a high voltage gain is required, the topology of Fig. 14.32(a) may be employed. Called "input offset storage," this approach incorporates two series capacitors at the input and places the

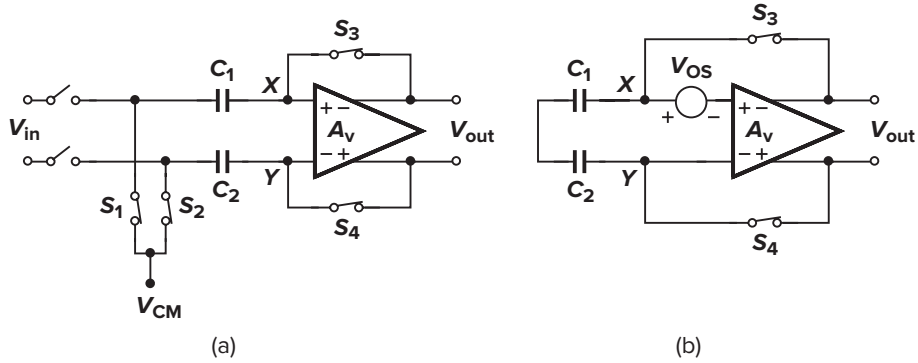


Figure 14.32 (a) Input offset storage; (b) circuit of (a) in the offset cancellation mode.

amplifier in a unity-gain negative-feedback loop during offset cancellation. Thus, from Fig. 14.32(b), $V_{out} = V_{XY}$ and $(V_{out} - V_{OS})(-A_v) = V_{out}$. That is

$$V_{out} = \frac{A_v}{1 + A_v} V_{OS} \tag{14.89}$$

$$\approx V_{OS} \tag{14.90}$$

In essence, the circuit reproduces the amplifier's offset at nodes X and Y, storing the result on C_1 and C_2. Note that for a zero differential input, the differential output is equal to V_{OS}. Therefore, the input-referred offset voltage of the overall circuit (after S_3 and S_4 turn off) equals V_{OS}/A_v if S_3 and S_4 match perfectly (and the input capacitance of the amplifier is much less than C_1 and C_2). In reality, however, when S_3 and S_4 turn off, their charge injection mismatch may saturate the amplifier if A_v is very large.

The general drawback of input and output storage techniques is that they introduce capacitors in the signal path, a particularly serious issue in op amps and feedback systems. The bottom-plate parasitic of the capacitors may reduce the magnitude of the poles in the circuit, thereby degrading the phase margin. Even in open-loop amplifiers, this parasitic may limit the settling speed, intensifying the speed-power trade-off.

To resolve the above issues, the offset cancellation scheme can isolate the signal path from the offset storage capacitors though the use of an "auxiliary" amplifier. Consider the topology shown in Fig. 14.33, where A_{aux} amplifies the differential voltage V_1 stored across C_1 and C_2 and subtracts the result from the output of A_1. We note that if $V_{OS1}A_1 = V_1A_{aux}$, then for $V_{in} = 0$, $V_{out} = 0$, and the circuit is free from offsets. The key point here is that C_1 and C_2 do not appear in the signal path.

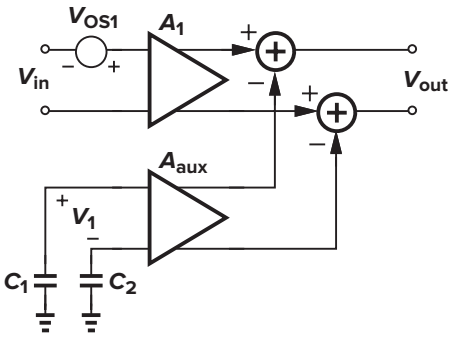


Figure 14.33 Addition of an auxiliary stage to remove the offset of an amplifier.

How is V_1 generated in Fig. 14.33? This is accomplished as illustrated in Fig. 14.34. Here, a second stage, A_2, is added and its output is sensed by A_{aux} during offset cancellation. To understand the operation, suppose that first only S_1 and S_2 are on, yielding $V_{out} = V_{OS1}A_1A_2$. Now, assume that S_3 and S_4 turn on, placing A_2 and A_{aux} in a negative-feedback loop. The reader can show that V_{out} then drops by a factor approximately equal to the loop gain: $V_{OS1}A_1A_2/(A_2A_{aux}) = V_{OS1}A_1/A_{aux}$. Stored across C_1 and C_2, this value is indeed the required V_1 in Fig. 14.33 because $(V_{OS1}A_1/A_{aux})A_{aux} = V_{OS1}A_1$.

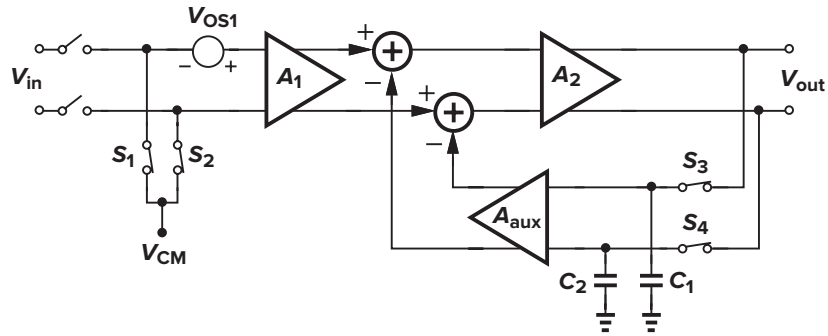


Figure 14.34 Auxiliary amplifier placed in a feedback loop during offset cancellation.

The topology of Fig. 14.34 suffers from two drawbacks. First, two voltage gain stages in the signal path may not be desirable in a high-speed op amp. Second, addition of the output voltages of A_1 and A_2 is quite difficult. For these reasons, the technique is usually realized as shown in Fig. 14.35(a), where each G_m stage is simply a differential pair and the R stage represents a transimpedance amplifier. As exemplified by Fig. 14.35(b), G_{m1} and R may in fact constitute a one-stage op amp, while G_{m2} adds an offset correction current at the low-impedance nodes X and Y.

Let us now examine the offset cancellation in Fig. 14.35(a) carefully, taking the offset voltage of G_{m2} into account as well. As depicted in Fig. 14.36, we can write

$$[G_{m1}V_{OS1} - G_{m2}(V_{out} - V_{OS2})]R = V_{out} \tag{14.91}$$

Thus,

$$V_{out} = \frac{G_{m1}RV_{OS1} + G_{m2}RV_{OS2}}{1 + G_{m2}R} \tag{14.92}$$

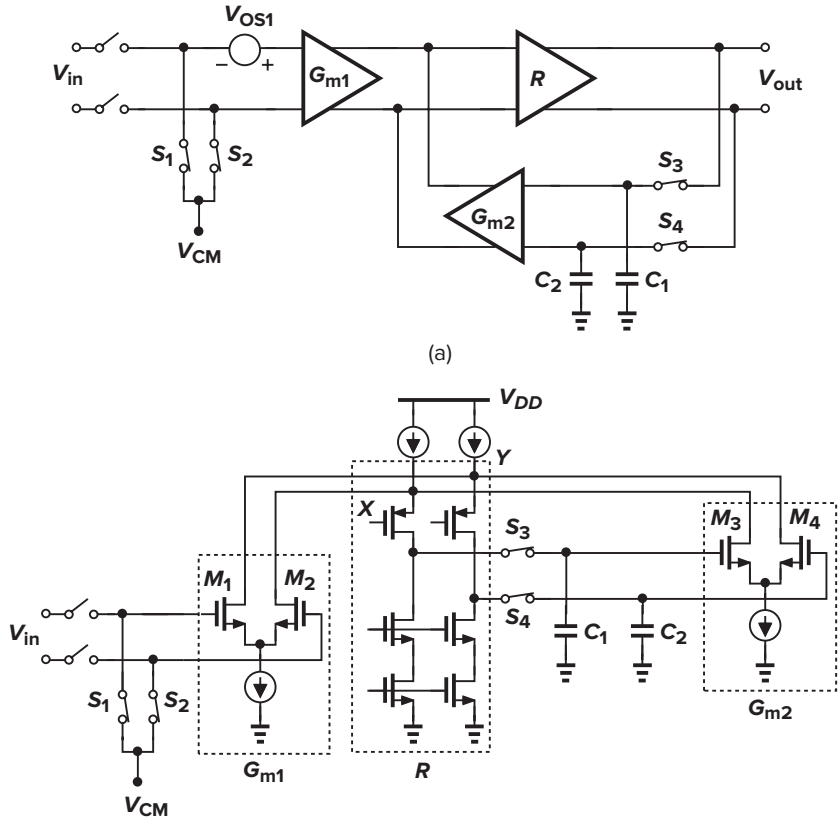


Figure 14.35 (a) Circuit of Fig. 14.34 using G_m and R stages; (b) realization of (a) in a folded-cascode op amp.

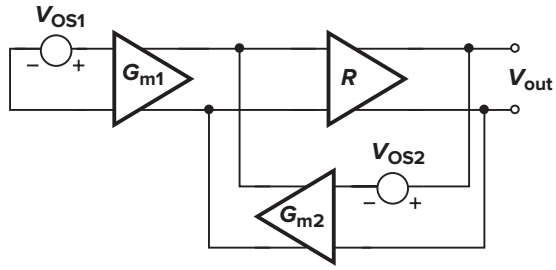


Figure 14.36 Circuit of Fig. 14.35(a) including offset of G_{m2}.

This voltage is stored on C_1 and C_2 after S_3 and S_4 turn off. The offset voltage referred to the main input is therefore given by

$$V_{OS,tot} = \frac{V_{out}}{G_{m1}R} \tag{14.93}$$

$$= \frac{V_{OS1}}{1 + G_{m2}R} + \frac{G_{m2}}{G_{m1}} \frac{V_{OS2}}{1 + G_{m2}R} \tag{14.94}$$

$$\approx \frac{V_{OS1}}{G_{m2}R} + \frac{V_{OS2}}{G_{m1}R} \tag{14.95}$$

where we have assumed that $G_{m2}R \gg 1$. If $G_{m2}R$ and $G_{m1}R$ are large, as in the op amp of Fig. 14.35(b), then $V_{OS,tot}$ is very small.

The offset cancellation of Fig. 14.35 warrants a cautionary note. Upon turning off, S_3 and S_4 may inject slightly unequal charges onto C_1 and C_2, respectively, creating an error voltage that is *not* corrected because the feedback loop is opened. The reader can prove that for a differential injection-induced error voltage of ΔV, the resulting input-referred offset voltage equals $(G_{m2}/G_{m1})\Delta V$. For this reason, G_{m2} is usually chosen to be on the order of $0.1G_{m1}$.

We should also mention that the unity-gain and precision multiply-by-two circuits described in Chapter 13 cancel the offset of the op amp as well. The proof is left to the reader.[3]

It is important to note that the offset cancellation techniques studied here require periodic refreshing because the junction and subthreshold leakage of the switches eventually corrupts the correction voltage stored across the capacitors. In a typical design, the offset must be refreshed at a rate of at least a few kilohertz.

14.2.3 Reduction of Noise by Offset Cancellation

Recall from previous sections that the offset of a differential amplifier can be viewed as a noise component having a very low frequency. We therefore expect that periodic offset cancellation can potentially reduce the (low-frequency) noise of the circuit as well.

Consider a simple differential amplifier that is to be used in the front end of a sampling system [Fig. 14.37(a)]. Here, the noise of A_1 directly corrupts V_{in}. The $1/f$ noise of A_1 proves especially problematic if the signal spectrum extends from zero to only a few megahertz, because the $1/f$ noise corner frequency is typically around 500 kHz to 1 MHz.

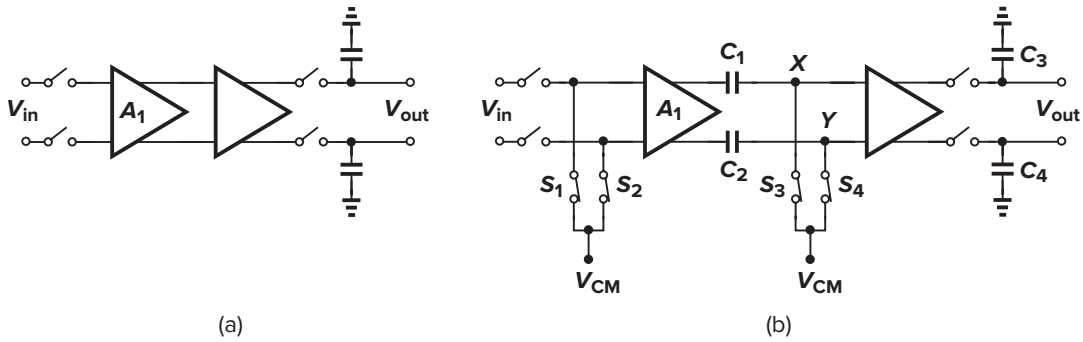


Figure 14.37 (a) Front end of a sampler; (b) circuit of (a) with offset cancellation applied to the first stage.

Now suppose the amplifier undergoes offset cancellation before *every* sampling operation [Fig. 14.37(b)]. That is, as depicted in Fig. 14.38, the input is disabled; the offset of A_1 is stored on C_1 and C_2; the input is enabled and amplified by A_1 and A_2 and stored on C_3 and C_4; and finally the sampling switches are turned off. How does the noise of A_1 affect the final output? Denoting the time elapsed from the end of offset cancellation to the end of sampling by $\Delta t = t_2 - t_1$, we recall that at $t = t_1$, $V_{XY} = 0$. Thus, from t_1 to t_2, only *high-frequency* noise components of A_1, on the order of $> 1/\Delta t$, change V_{XY} significantly. In other words, offset cancellation suppresses noise frequencies below roughly $1/\Delta t$.

To better understand this concept, let us consider a numerical example. Assuming that $\Delta t = 10$ ns, we examine two noise components, one at 1 MHz and another at 10 MHz, approximating each with a sinusoid (Fig. 14.39). For a sinusoid of amplitude A and frequency f, the maximum slew rate is equal to $2\pi f A$,

[3] If, as shown in Fig. 13.35, an equalizing switch is added to the circuit, then the op amp offset may not be removed.

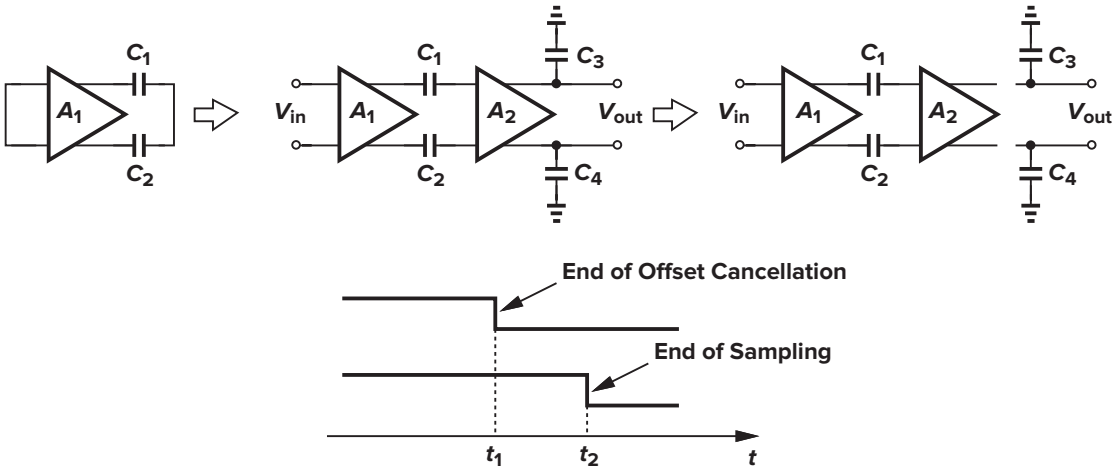


Figure 14.38 Sequence of operations in the sampler.

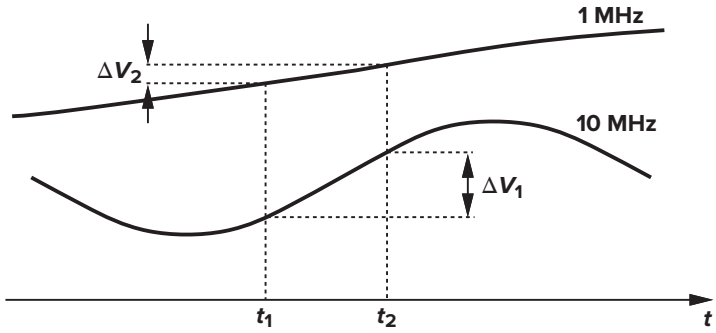


Figure 14.39 Variation of 1-MHz and 10-MHz noise components in a time interval of 10 ns.

and hence the maximum variation in Δt seconds is $2\pi f A \Delta t$. Normalizing this value to the amplitude, we obtain the change for the 1-MHz and 10-MHz components as $\Delta V_1/A = 6.3\%$ and $\Delta V_2/A = 63\%$, respectively. We therefore conclude that noise frequencies below a few megahertz do not have sufficient time to change if the sampling occurs only 10 ns after the end of offset cancellation.

Originally utilized in charge-coupled devices (CCDs), the foregoing property of offset cancellation is called "correlated double sampling" (CDS) because it involves two consecutive sampling operations (the first being offset storage) that are so tightly spaced in time that they do not allow (low-frequency) noise components to vary significantly. A powerful technique, CDS finds wide usage in suppressing the $1/f$ noise of MOS circuits. Nonetheless, it leads to aliasing of wideband noise [5].

14.2.4 Alternative Definition of CMRR

Recall from Chapter 4 that common-mode rejection is represented by the change in the differential output divided by the change in the input common-mode level, and the CMRR is defined as the differential gain divided by this quantity. We also noted that in fully differential circuits, the finite output impedance of the tail current source and asymmetries limit the common-mode rejection.

Now consider a differential circuit sensing an input CM change, $\Delta V_{in,CM}$. If the differential output voltage changes by ΔV_{out} while the differential input voltage is zero, we can say that the output *offset* voltage of the circuit has changed by ΔV_{out}. In other words, common-mode rejection can be viewed as the change in the output offset divided by the change in the input CM level. Following the notation in

Chapter 4, we write

$$A_{CM-DM} = \frac{\Delta V_{OS,out}}{\Delta V_{CM,in}} \tag{14.96}$$

Since CMRR $= A_{DM}/A_{CM-DM}$, we have

$$\text{CMRR} = \frac{A_{DM}}{\dfrac{\Delta V_{OS,out}}{\Delta V_{CM,in}}} \tag{14.97}$$

$$= \frac{\dfrac{\Delta V_{CM,in}}{\Delta V_{OS,out}}}{A_{DM}} \tag{14.98}$$

Noting that $\Delta V_{OS,out}/A_{DM}$ is in fact the input-referred offset voltage, we have

$$\text{CMRR} = \frac{\Delta V_{CM,in}}{\Delta V_{OS,in}} \tag{14.99}$$

The above result proves useful in analyzing the behavior of circuits. For example, suppose an op amp incorporates a PMOS differential pair at the input. Which one of the topologies shown in Fig. 14.40 yields a higher CMRR? In Fig. 14.40(a), body effect is eliminated and the threshold voltages of M_1 and M_2 are independent of the input CM level. In Fig. 14.40(b), on the other hand, M_1 and M_2 experience body effect and if they suffer from mismatches in their body effect coefficients, then the difference between V_{TH1} and V_{TH2}, i.e., the input offset voltage, *varies* with the input CM level, degrading the common-mode rejection.

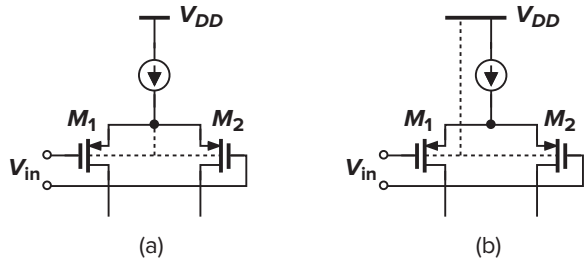


Figure 14.40 PMOS differential pair (a) without and (b) with body effect.

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Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. Also, assume that all transistors are in saturation.

14.1. The input-output characteristic of an amplifier is approximated as $y(t) = \alpha_1 x(t) + \alpha_2 x^2(t)$ in the range $x = [0 \ \ x_{max}]$.
(a) What is the maximum nonlinearity?
(b) What is the THD for $x(t) = (x_{max} \cos \omega t + x_{max})/2$.

14.2. In the circuits of Fig. 14.6, $W/L = 20/0.5$ and $I = 0.5$ mA. Calculate the harmonic distortion in each circuit if the input signal has a peak amplitude of 100 mV. How do the results change if we double W/L or I?

14.3. For the circuits of Fig. 14.6, plot the THD and the input-referred thermal noise as a function of **(a)** W/L, **(b)** I. Identify the trade-offs among noise, linearity, and power dissipation.

14.4. In Fig. 14.6, *two* effects lead to a trade-off between nonlinearity and voltage gain. Describe these effects.

14.5. The circuit of Fig. 14.6(a) is designed with $W/L = 50/0.5$, $I = 1$ mA, and $R_D = 2$ kΩ. The circuit is placed in a feedback loop similar to that of Fig. 14.7 with $\beta = 0.2$ and senses an input sinusoid with a peak amplitude of 10 mV. Calculate the THD at the output.

14.6. Suppose that in Fig. 14.16, A_1 and A_2 have an input-referred noise voltage V_n. Neglecting other sources of noise, calculate the input-referred noise voltage of the overall circuit.

14.7. Equation 14.36 suggests that if the open-loop gain, α_1, increases while other parameters remain constant, then the harmonic distortion drops sharply. Repeat Problem 14.14.5 with $W/L = 200/0.5$ to achieve a higher open-loop gain and explain the results.

14.8. Equation 14.36 suggests that if $\beta\alpha_1 \gg 1$, then $b/a \propto \beta^{-2}$. Repeat Problem 14.14.5 with $\beta = 0.4$.

14.9. Suppose the nonlinear feedforward amplifier in Fig. 14.7 is characterized by $y(t) = \alpha_1 x(t) + \alpha_3 x^3(t)$. Estimate the magnitude of the third harmonic at the output of the overall system.

14.10. As mentioned in Chapter 2, MOS devices operating in the subthreshold region exhibit an exponential behavior: $I_D = I_0 \exp[V_{GS}/(\zeta V_T)]$. Suppose both of the circuits shown in Fig. 14.6 operate in the subthreshold region. Derive expressions for the harmonic amplitudes if the input signal is much less than ζV_T. For the differential pair, first prove that $I_{D1} - I_{D2} \propto \tanh[V_{in}/(2\zeta V_T)]$ and then write the Taylor expansion of the hyperbolic tangent.

14.11. The mobility of MOSFETs is in fact a function of the gate-source voltage and expressed as $\mu = \mu_0/[1 + \theta(V_{GS} - V_{TH})]$, where θ is an empirical factor (Chapter 17). Assuming that $\theta(V_{GS} - V_{TH}) \ll 1$ and using the relationship $(1 + \epsilon)^{-1} \approx 1 - \epsilon$ for $\epsilon \ll 1$, calculate the third harmonic in the circuit of Fig. 14.6(a).

14.12. The input devices of a differential pair have an effective length of 0.5 μm.
(a) Assuming that $\Delta V_{TH} = 0.1 t_{ox}/\sqrt{WL}$ and neglecting other mismatches, determine the minimum width of the transistors such that $V_{OS} \leq 5$ mV.
(b) If the tail current is 1 mA, what is the maximum input swing that gives a THD of 1%?

14.13. Repeat Problem 14.14.12 if the tolerable input offset is 2 mV and compare the results.

14.14. Determine the dimensions of M_1 and M_2 in Fig. 14.28 such that $I_{D1} \approx I_{D2} = 0.5$ mA, $\Delta I_D/I_D = 2\%$, and $V_{GS} - V_{TH} = 0.5$ V. Assume that $\Delta V_{TH} = 0.1 t_{ox}/\sqrt{WL}$ and neglect other mismatches.

14.15. Source degeneration can improve the matching between current sources if resistor mismatches are small. Prove that in the circuit of Fig. 14.41,

$$\frac{\Delta I_D}{I_D} = \frac{1}{1 + g_m R_S} \left[\frac{\Delta(\mu_n C_{ox})}{\mu_n C_{ox}} + \frac{\Delta(W/L)}{(W/L)} - \frac{2\Delta V_{TH}}{V_{GS} - V_{TH}} - g_m \Delta R_S \right] \tag{14.100}$$

where ΔR_S denotes the mismatch between R_{S1} and R_{S2}. Note that for an appreciable reduction of $\Delta I/I_D$, R_S must be greater than $1/g_m$.

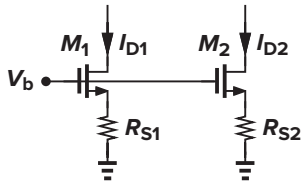


Figure 14.41

14.16. In the circuit of Fig. 14.29, assume that $\alpha_j = \beta_j$ but $x_1(t) = A\cos\omega t$ and $x_2(t) = A\cos(\omega t + \theta)$, where θ denotes a small phase mismatch. Calculate the magnitude of the second harmonic at the output.

14.17. In the circuit of Fig. 14.42, M_3 and M_4 suffer from a threshold mismatch of ΔV_{TH} and the circuit is otherwise symmetric. Assuming that $\lambda \neq 0$ but $\gamma = 0$, calculate the input-referred offset voltage. What happens as $R_D \to \infty$?

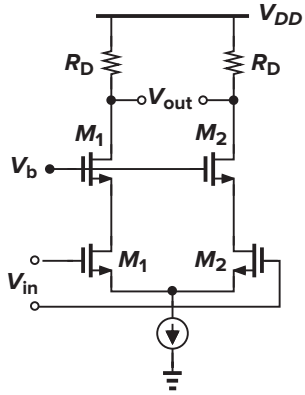


Figure 14.42

14.18. In the circuit of Fig. 14.32, the amplifier has an input capacitance (between X and Y) equal to C_{in}. Calculate the input offset voltage after offset compensation.

14.19. The circuit of Fig. 14.32 is designed for an input offset voltage of 1 mV. If the width of the transistors in the input differential pair of the amplifier is doubled, what is the overall input offset voltage? (Neglect the input capacitance of the amplifier.)

14.20. Explain why the circuit of Fig. 14.27 suffers from a trade-off between the input offset and the output voltage swing (for a given tail current).